



A comprehensive experimental and kinetic modeling study of *n*-propylbenzene combustion



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ABSTRACT

This work presents a comprehensive experimental and kinetic modeling study on the combustion of *n*-propylbenzene. Flow reactor pyrolysis of *n*-propylbenzene at 0.04, 0.2 and 1 atm and laminar premixed flames of *n*-propylbenzene at 0.04 atm with equivalence ratios of 0.75 and 1.00 were investigated with synchrotron vacuum ultraviolet photoionization mass spectrometry. Jet stirred reactor (JSR) oxidation of *n*-propylbenzene at 10 atm with equivalence ratios of 0.5, 1.0, 1.5 and 2.0 was investigated with gas chromatography. A detailed kinetic model for *n*-propylbenzene combustion with 340 species and 2069 reactions was developed and validated against the data measured in this work. Model analyses such as rate of production analysis and sensitivity analysis were also performed to reveal the key pathways in the consumption of fuel and formation of polycyclic aromatic hydrocarbons (PAHs). The analysis results demonstrate that the benzylic C–C bond dissociation reaction is crucial for the decomposition of *n*-propylbenzene in the pyrolysis and rich flame. Low temperature oxidation reactions play important roles in the high pressure JSR oxidation of *n*-propylbenzene. In addition, the formation pathways of PAHs are strongly related to the fuel structure, especially for the formation of bicyclic PAHs such as indene and naphthalene. Furthermore, the present model was also validated against previous experimental data of *n*-propylbenzene combustion under a wide range of conditions, including ignition delay times, laminar flame speeds, extinction strain rates, speciation profiles in atmospheric pressure JSR oxidation, flow reactor oxidation and high pressure shock tube pyrolysis and oxidation.

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1. Introduction

The alkylbenzene chemical class is an important component of gasoline, diesel oils and jet fuels, and constitutes a great part in their surrogate fuels [1–3]. The combustion of alkylbenzenes tends to produce polycyclic aromatic hydrocarbons (PAHs) and soot [4,5] which can cause severe environmental and health problems

[6]. Among alkylbenzenes, *n*-propylbenzene is a typical compound with moderate molecular weight and alkyl-chain length, thus it is widely used as a surrogate component for jet fuels [1,7–9] and diesel oils [3].

There are some previous experimental investigations on *n*-propylbenzene combustion including both speciation and global parameter measurements. The very early work of Robaugh et al. [10] investigated the very low-pressure pyrolysis of *n*-propylbenzene at near 1000 K and determined the rate constants of *n*-propylbenzene bond dissociation reactions. Litzinger et al. [11] measured stable products in the flow reactor oxidation of *n*-propylbenzene at 1 atm and near 1060 K using gas chromatography (GC). Anderson et al. [12] investigated the coflow diffusion flame of methane doped with *n*-propylbenzene at 1 atm, which concerns on the formation of polycyclic aromatic hydrocarbons (PAHs). Dagaut et al. [13] studied the jet stirred reactor (JSR) oxidation of *n*-propylbenzene at 1 atm, 900–1250 K and equivalence

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ratios (ϕ) from 0.5 to 1.5. Gudiyaella and Brezinsky investigated the high pressure oxidation (25, 50 atm) [14] and pyrolysis (50 atm) [15] of *n*-propylbenzene in a high pressure single pulse shock tube. Wang et al. [16] investigated the rich laminar premixed flame of *n*-propylbenzene at 0.04 atm. Global combustion parameters were extensively measured under a wide range of conditions for *n*-propylbenzene, such as ignition delay times [17–19], laminar flame speeds [20–23] and extinction strain rates [20–22,24].

There are also some theoretical investigations on elementary reactions in *n*-propylbenzene combustion. Robinson and Lindstedt [25] calculated rate constants for H-atom abstraction reactions of *n*-propylbenzene by H and CH₃. Altarawneh and Dlugogorski [26] calculated reactions of HO₂ with *n*-propylbenzene and the phenylpropyl radicals. Zhou et al. [27] calculated rate constant for the H-atom abstraction reaction by O₂ on the benzylic site. Some kinetic models of *n*-propylbenzene were also proposed by previous researchers. Litzinger et al. [11] proposed a kinetic model for *n*-propylbenzene oxidation based on the validation against their flow reactor experimental data. Dagaut et al. [13] proposed a detailed kinetic model for *n*-propylbenzene oxidation validated against their JSR experimental data. Darcy et al. [18,19] proposed a detailed kinetic model for *n*-propylbenzene oxidation validated against their ignition delay time data and previous JSR oxidation data. Gudiyaella and Brezinsky [15] proposed a detailed kinetic model for *n*-propylbenzene validated against their shock tube pyrolysis and oxidation data and previous JSR oxidation data.

It should be noted that the pyrolysis experiments and high pressure oxidation experiments are still insufficient and most of previous speciation experiments are limited in detection of radicals and other reactive intermediates. Furthermore, most of previous experimental and kinetic modeling studies focused on the decomposition and oxidation kinetics of *n*-propylbenzene, while few concerned the formation of PAHs. Finally, a comprehensive kinetic model for *n*-propylbenzene which is developed with reliable rate constants and validated against a vast amount of experimental data of *n*-propylbenzene combustion is still lacking.

The first target of this work is to report an experimental investigation on *n*-propylbenzene combustion under a variety of reaction circumstances and pressures. Flow reactor pyrolysis of *n*-propylbenzene at various pressures (0.04, 0.2 and 1 atm) and laminar premixed flames of *n*-propylbenzene at 0.04 atm ($\phi = 0.75$ and 1.00) were investigated with synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS). Jet stirred reactor (JSR) oxidation of *n*-propylbenzene at 10 atm ($\phi = 0.50$, 1.00, 1.50 and 2.00) was studied with GC. The second target of this work is to report a comprehensive kinetic model for *n*-propylbenzene pyrolysis and oxidation. The model incorporated low-to-high temperature oxidation reactions and PAH formation reactions and was validated against experimental data for *n*-propylbenzene combustion obtained in this work and literature experimental data. Moreover, detailed modeling analyses were performed to provide insights into the chemistry of *n*-propylbenzene consumption and PAH formation.

2. Experimental section

2.1. Flow reactor pyrolysis

The flow reactor pyrolysis experiments were performed at the National Synchrotron Radiation Laboratory in Hefei, China. Detailed descriptions of the beamlines and pyrolysis apparatus were presented in our previous work [28–32]. *n*-Propylbenzene with a purity of $\geq 99\%$ was purchased from Sinopharm Chemical Reagent Limited Co., Shanghai, China. It was gasified in a vaporizer with the temperature maintaining at 475 K which is about 40 K higher than its boiling point. After vaporization, the gas mixture

Table 1

Conditions of the JSR oxidation of *n*-propylbenzene.

ϕ	<i>P</i> (atm)	Fuel (%)	O ₂ (%)	N ₂ (%)	τ (s)
0.50	10	0.1	2.4	97.5	0.5
1.00	10	0.1	1.2	98.7	0.5
1.50	10	0.1	0.8	99.1	0.5
2.00	10	0.1	0.6	99.3	0.5

of *n*-propylbenzene (1%) and argon (99%) with a total flow rate of 1000 standard cubic centimeters per minute (sccm) was fed into the flow tube in the reactor. The flow tube is made of α -alumina to reduce wall catalytic effects [33–35] and has a 150 mm heating length. A small inner diameter (7.0 mm) is used to ensure strong radial diffusion effects, reduce radial concentration gradients and achieve adequately homogeneous reaction circumstances according to the experiences in previous laminar flow reactor experiments [36–38]. The temperature distributions along the centerline of the flow tube were measured by a S-type thermocouple. Detailed descriptions of the temperature measurement method were provided in previous studies [30,31]. Methodologies for intermediate identification and mole fraction evaluation were also reported in detail previously [31]. The uncertainties of evaluated mole fractions were estimated to be within $\pm 25\%$ for pyrolysis products with known photoionization cross sections (PICSS) and a factor of 2 for those with estimated PICSS. The PICSSs of pyrolysis species are available in the online database [39].

2.2. Jet stirred reactor oxidation

The JSR oxidation experiments were performed at C.N.R.S in Orléans, France. Table 1 lists the detailed experimental conditions. The jet stirred reactor used in this work and the experimental procedure are similar to those used earlier [40]. It consists of a small fused silica sphere of 40 mm outer diameter (30.5 cm³). Four nozzles of 1 mm inner diameter were mounted in the sphere to feed the gases and achieve stirring. Two regulated heating wires of ca. 1.5 kW maintained the temperature of the reactor at the desired working temperature. Good thermal homogeneity along the whole vertical axis of the reactor (gradients of ca. 1 K/cm) was observed for each experiment by thermocouple (0.16 mm chromel–alumel) measurements (typical change ≤ 5 K). A high-pressure liquid chromatography pump (Hewlett Packard 1100) was used to deliver sonically degassed *n*-propylbenzene ($> 98\%$ purity) mixture to an atomizer–vaporizer. A high degree of dilution (0.1% volume of fuel) was used in order to reduce temperature gradients in the reactor caused by the oxidation heat release. High-purity oxygen (99.995% pure) was used in the experiment. The reactants were diluted by nitrogen (< 50 ppm O₂, < 1000 ppm Ar, < 5 ppm H₂) and mixed at the entrance of the injectors. All the gases were preheated before injection to minimize temperature gradients inside the reactor. The reacting mixtures were sampled by means of a fused silica low-pressure sonic probe. The samples were analyzed online by GC-mass spectrometry (MS)-flame ionization detector (FID) and off-line after collection and storage in 1 L Pyrex bulbs. Off-line analyses were performed using GC equipped with capillary columns (DB-5 ms, Al₂O₃-KCl, Poraplot U and Carboplot-P7), a thermal conductivity detector (TCD), and an FID.

2.3. Laminar premixed flame

The laminar premixed flame experiments were performed at National Synchrotron Radiation Laboratory in Hefei, China. Detailed descriptions of the flame apparatus were published elsewhere [28,29]. The flame conditions are listed in Table 2. The flame was stabilized on a McKenna burner with a diameter of 6.0 cm. *n*-Propylbenzene was gasified in a vaporizer with the tem-

Table 2
Conditions of the laminar premixed flames of *n*-propylbenzene.

ϕ	P (atm)	Fuel (%)	O ₂ (%)	Ar (%)	V (cm s ^{-1 a})
0.75	0.04	2.94	47.06	50	50
1.00	0.04	3.85	46.15	50	50
1.79 ^b	0.04	6.49	43.51	50	50

^a V is the flow velocity of inlet mixture at 300 K.

^b Reported in our previous work [16].

perature maintaining at 475 K. All measurements were performed along the centerline of the burner. The flame temperature profiles were measured using a 0.10-mm-diameter Pt-6%Rh/Pt-30%Rh thermocouple coated with Y₂O₃-BeO anti-catalytic ceramic to inhibit the catalytic effects [41]. The temperatures were corrected for radiative heat loss [42] and cooling effects of the sampling nozzle [43]. The uncertainties of the maximum flame temperatures were estimated to be within ± 100 K. Methodologies of intermediate identification and mole fraction evaluation were reported in detail previously [28,44]. The uncertainties of evaluated mole fractions were estimated to be within $\pm 10\%$ for major species, $\pm 25\%$ for intermediates with accurately known PICS, and a factor of 2 for those with estimated PICS [28].

3. Kinetic model

A detailed kinetic model for *n*-propylbenzene combustion consisting of 340 species and 2069 reactions was developed based on our recent comprehensive ethylbenzene model [45] which has been widely validated against combustion data of ethylbenzene, styrene, toluene, benzene, and cyclopentadiene [45–47]. The sub-mechanism of *n*-propylbenzene was developed in this work. The unimolecular decomposition reactions of the fuel and fuel radicals are one of the significant high temperature reaction classes of *n*-propylbenzene [13,48,49]. Among the unimolecular decomposition reactions, benzylic C–C bond scission reaction is the most favored one due to the rather low bond dissociation energy (BDE) of benzylic C–C bond [50]. In the present model, the unimolecular dissociation reactions of *n*-propylbenzene were referred to those of ethylbenzene, since the BDEs of *n*-propylbenzene are very similar to the similar bonds in ethylbenzene, as shown in Fig. S1 in the *Supplementary materials* [51]. Recently, Ye et al. [52] theoretically calculated rate constants for the unimolecular decomposition reactions of ethylbenzene. In the present model, the rate constants of unimolecular C–C and C–H bond decomposition reaction of *n*-propylbenzene were referred to those of ethylbenzene calculated by Ye et al. [52]. Furthermore, the unimolecular rearrangement reactions of phenylalkyl radicals with relatively long alkyl chain belong to the isomerization of fuel radical which is a typical reaction class. In the present model, the 1,2-phenyl migration reaction of 1-phenyl-prop-2-yl radical was incorporated and the rate constant was taken from the theoretical calculation results of Wang et al. [53].

The H-atom abstraction reactions on the fuel are another significant high temperature reaction class. In the present model, the rate constants of H-atom abstraction reactions of *n*-propylbenzene by H and CH₃ were taken from the theoretical calculation results of Robinson and Lindstedt [25]. For the H-atom abstraction reactions of *n*-propylbenzene by OH and HO₂, the rate constant at the benzylic site was referred to the analogous reaction of ethylbenzene, while the rate constants at the other two sites on the alkyl chain were referred to the analogous reactions of *n*-butane by considering the differences in degeneracy. The rate constant of H-atom abstraction reaction by O₂ at the benzylic site was taken from the theoretical calculation results of Zhou et al. [27].

Low temperature reaction classes were also included in the sub-mechanism of *n*-propylbenzene with the rate constants referred to

the analogous reactions of toluene [46], ethylbenzene [45], alkanes and alkenes [54,55]. The rate constant of the O₂ addition reactions of the 1-phenylpropyl radical was referred to that of benzyl [56], while the rate constants of the O₂ addition reactions of the 1-phenyl-2-propyl radical and the 1-phenyl-3-propyl radical were referred to those of the *n*-propyl radical [57]. The rate constants of the concerted elimination reactions of phenylpropylperoxy radicals were referred to those of phenylethylperoxy radicals that were calculated by Altarawneh et al. [58]. The rate constants for the isomerization reactions of phenylpropylperoxy radicals to form hydroperoxyphenylalkyl radicals (QOOH) were referred to those of hexenylperoxy radicals [55]. The rate constants for the O₂ addition reactions of QOOH to produce phenylalkylhydroperoxy radicals (O₂QOOH), the consumption reactions of O₂QOOH to produce ketohydroperoxides and the decomposition reactions of ketohydroperoxides were taken from the work of Darcy et al. [19].

The thermodynamic data of *n*-propylbenzene were taken from the database of Goos et al. [59], while the thermodynamic data of phenylpropyl radicals were taken from the work of Darcy et al. [19]. The thermodynamic data of phenylpropene isomers and the low temperature oxidation products of *n*-propylbenzene were estimated using the THERGAS software [60]. The transport data of the species in the sub-mechanism of *n*-propylbenzene were taken from the work of Wang et al. [16]. The reaction mechanism, thermodynamic data and transport data are provided in the *Supplementary materials*. The simulations in this work were performed with the Chemkin-Pro software [61].

4. Results and discussion

4.1. Flow reactor pyrolysis

Dozens of pyrolysis species, such as major species, free radicals, isomers and PAHs, were detected in the pyrolysis of *n*-propylbenzene (A1C₃H₇). The comparison of experimental and simulated mole fraction profiles of reactants and products are illustrated in Figs. 1, 2 and S2 in the *Supplementary materials*. Toluene (A1CH₃), styrene (A1C₂H₃), ethylbenzene (A1C₂H₅) and benzene (A1) were found as abundant aromatic products smaller than *n*-propylbenzene at all pressures. Benzyl radical (A1CH₂) was also significantly produced at low pressure with the maximum mole fraction of 1.2×10^{-3} , however, at higher pressure, its concentration became too low to be detected. Among the PAH species, bibenzyl (C₁₄H₁₄), diphenylmethane (C₁₃H₁₂) and indene (C₉H₈) were significantly produced. Furthermore, ethylene (C₂H₄), acetylene (C₂H₂), methane (CH₄) and H₂ became the most abundant smaller products. The simulation was carried out with the Plug Flow Reactor module in the Chemkin-Pro software [61] using the measured centerline temperature profiles as input parameters.

The experimental and simulated mole fraction profiles of the decomposition products of *n*-propylbenzene are shown in Figs. 1 and S2. Rate of production (ROP) analysis was performed to investigate the main decomposition pathways of *n*-propylbenzene under pyrolysis conditions. Similar to ethylbenzene [45], the high temperature decomposition of *n*-propylbenzene can proceed through several pathways, such as unimolecular decomposition, H-atom abstraction, ipso-substitution and so on, among which unimolecular C–C bond dissociation and H-atom abstraction are the most favored pathways. Among all bonds in the *n*-propylbenzene molecule, the benzylic C–C bond (BDE = 71.7 kcal/mol) is the weakest [51] and its dissociation can produce the benzyl radical and the ethyl radical which were both detected in the 0.04 atm experiment. The ROP analysis shows that over 70% of *n*-propylbenzene undergoes the benzylic C–C bond dissociation reaction at 0.04 atm. In contrast, the dissociation reactions of C–H bonds, phenylic C–C bond and alkylic C–C bond only have

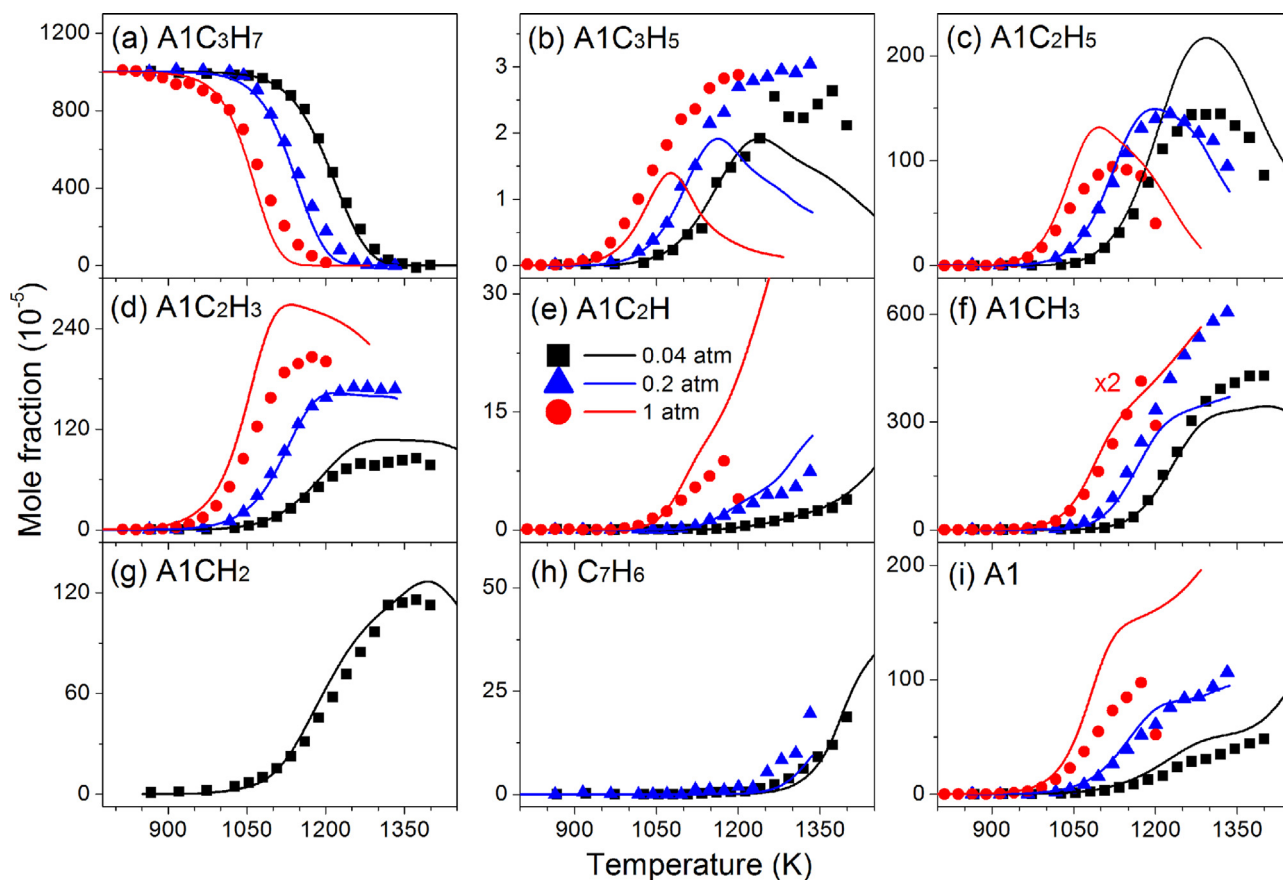


Fig. 1. Experimental (symbols) and simulated (lines) mole fraction profiles of *n*-propylbenzene ($A1C_3H_7$), phenylpropene isomers ($A1C_3H_5$), styrene ($A1C_2H_3$), phenylacetylene ($A1C_2H$), toluene ($A1CH_3$), benzyl radical ($A1CH_2$), fulvenallene (C_7H_6) and benzene ($A1$) in the pyrolysis of *n*-propylbenzene at 0.04, 0.2 and 1 atm.

negligible contributions to the consumption of *n*-propylbenzene due to their much higher BDEs. The rest of *n*-propylbenzene is mainly consumed by the H-atom abstraction reactions by H (22%) and CH_3 (2%) or the ipso-substitution reaction by H (4%).

As the pressure increases from 0.04 to 1 atm, the contribution of benzylic C–C bond dissociation decreases (46%) while that of H-atom abstraction increases (32%). The same phenomenon was also observed in our previous work on toluene and ethylbenzene pyrolysis [45,46]. The experimental and simulated mole fraction profiles of *n*-propylbenzene are further compared with the results of ethylbenzene in its pyrolysis under same conditions [45], which is shown in Fig. 3. It can be seen from Fig. 3 that at both 0.04 and 1 atm, *n*-propylbenzene has lower decomposition temperatures than ethylbenzene. This is mainly because the BDE of benzylic C–C bond in *n*-propylbenzene is 2.1 kcal/mol weaker than that in ethylbenzene [51]. In addition, *n*-propylbenzene has another alkylic C–C bond which has a relatively low BDE (84.2 kcal/mol) [51].

The H-atom abstraction reactions of *n*-propylbenzene mainly occur on the alkyl chain, producing three phenylpropyl radicals, i.e. 1-phenylpropyl radical ($A1\bullet CHCH_2CH_3$), 1-phenyl-prop-2-yl radical ($A1CH_2\bullet CHCH_3$) and 1-phenyl-prop-3-yl radical ($A1CH_2CH_2\bullet CH_3$). 1-Phenylpropyl and 1-phenyl-prop-3-yl radicals mainly suffer β -C–C scission reactions, producing styrene + methyl radical and benzyl radical + ethylene, respectively. In contrast, the 1-phenyl-prop-2-yl radical mainly undergoes 1,2-phenyl migration reaction, producing the 2-phenylpropyl radical ($A1CH(CH_3)\bullet CH_2$) which subsequently decomposes to styrene and the methyl radical. Thus it can be concluded that benzyl radical and styrene are the most significant intermediates in the pyrolysis of *n*-propylbenzene. The production of styrene has a remarkable increase from low pressure

to atmospheric pressure, which indicates that the contribution of H-atom abstraction reactions increases as pressure increases.

Benzyl radical can easily convert to toluene and ethylbenzene by adding an H atom and a methyl radical, which are also the dominant formation pathways of toluene and ethylbenzene, respectively. In addition, benzyl radical can be consumed by the self-combination reaction producing bibenzyl and by the unimolecular decomposition reactions producing fulvenallene, benzyne ($o\text{-}C_6H_4$) and cyclopentadienyl radical (C_5H_5). Styrene is mainly consumed by the H-atom abstraction reactions producing 1-phenylvinyl and 2-phenylvinyl radicals which both decompose to phenylacetylene ($A1C_2H$). Other important consumption pathways of styrene include the ipso-substitution reaction by H atom producing benzene and the vinyl radical, the H-attack reaction producing phenyl radical ($A1\bullet$) and ethylene, and the unimolecular decomposition reaction producing benzene and vinylidene. Detailed discussion of the decomposition pathways of the products smaller than the benzyl radical and styrene has been introduced in our previous studies [45,46], thus will not be repeated herein. Besides, the production of many small hydrocarbon products is highly dependent on the decomposition of the benzyl radical, such as C_3 and C_2 species. At 1 atm, most of benzyl radical is consumed by the self-combination reaction or the combination reactions with H and CH_3 , while only little is consumed by the decomposition reaction. Therefore the formation of these small hydrocarbon species is suspended at 1 atm.

Figure 2 illustrates the experimental and simulated mole fraction profiles of nine typical PAHs in the pyrolysis of *n*-propylbenzene. It can be observed from Fig. 2 that the most abundantly produced PAHs are bibenzyl and diphenylmethane

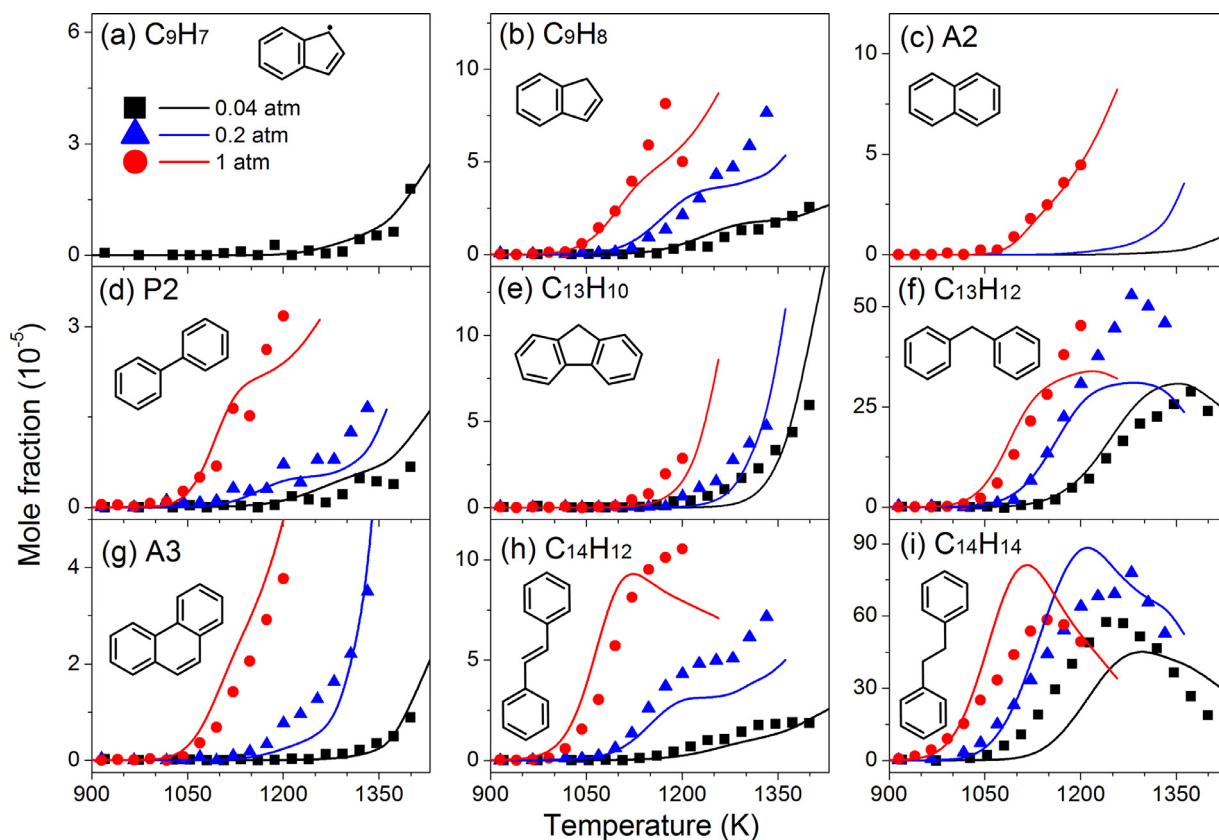


Fig. 2. Experimental (symbols) and simulated (lines) mole fraction profiles of indenyl (C_9H_7), indene (C_9H_8), naphthalene (A2), biphenyl (P2), fluorene ($C_{13}H_{10}$), diphenylmethane ($C_{13}H_{12}$), phenanthrene (A3), stilbene ($C_{14}H_{12}$), and bibenzyl ($C_{14}H_{14}$) in the pyrolysis of *n*-propylbenzene at 0.04, 0.2 and 1 atm.

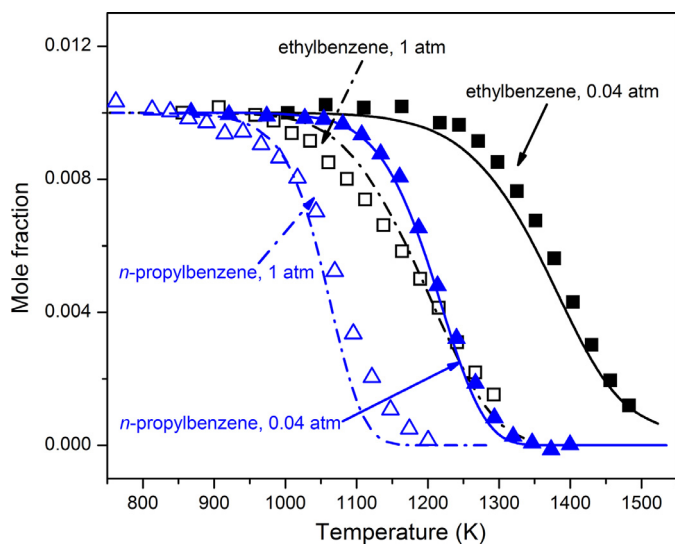


Fig. 3. Comparison of experimental and simulated mole fraction profiles of ethylbenzene and *n*-propylbenzene at 0.04 and 1 atm.

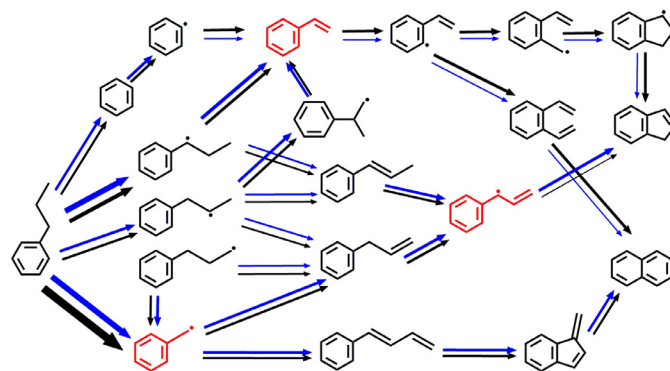


Fig. 4. Main formation pathways of indene and naphthalene in the pyrolysis of *n*-propylbenzene at 0.04 atm (black) and 1 atm (blue). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

($C_{13}H_{12}$), both of which are non-fused PAHs. The two PAHs are mainly formed from the self-combination of benzyl and the combination of the benzyl radical and the phenyl radical, respectively. Similarly, the formation of biphenyl (P2) is also controlled by the self-combination of the phenyl radical and the combination of the phenyl radical and benzene. The consumption of bibenzyl and diphenylmethane is controlled by dehydrogenation reactions and cyclization reactions, forming fused PAHs such

as phenanthrene and fluorene, which have been discussed in our previous work [46]. Figure 4 shows the main formation pathways of indene and naphthalene which are two important bicyclic PAHs. For indene, the cyclization and subsequent dehydrogenation reactions of 1-phenyl-propen-2-yl ($A1CH\bullet CCH_3$) and 1-phenyl-propen-3-yl ($A1CHCH\bullet CH_2$) radicals are its main formation pathways, especially at 1 atm. 1-Phenyl-propen-2-yl and 1-phenyl-propen-3-yl radicals are the H-atom abstraction reaction products of 1-phenyl-prop-2-ene and 1-phenyl-prop-3-ene, which are mainly produced from the β -C-H scission of phenylpropyl radicals. In addition, phenylpropenyl radicals can also be produced from the combination of benzyl and acetylene, as shown in Fig. 4. At 0.04 atm, indene can also be produced from the reaction

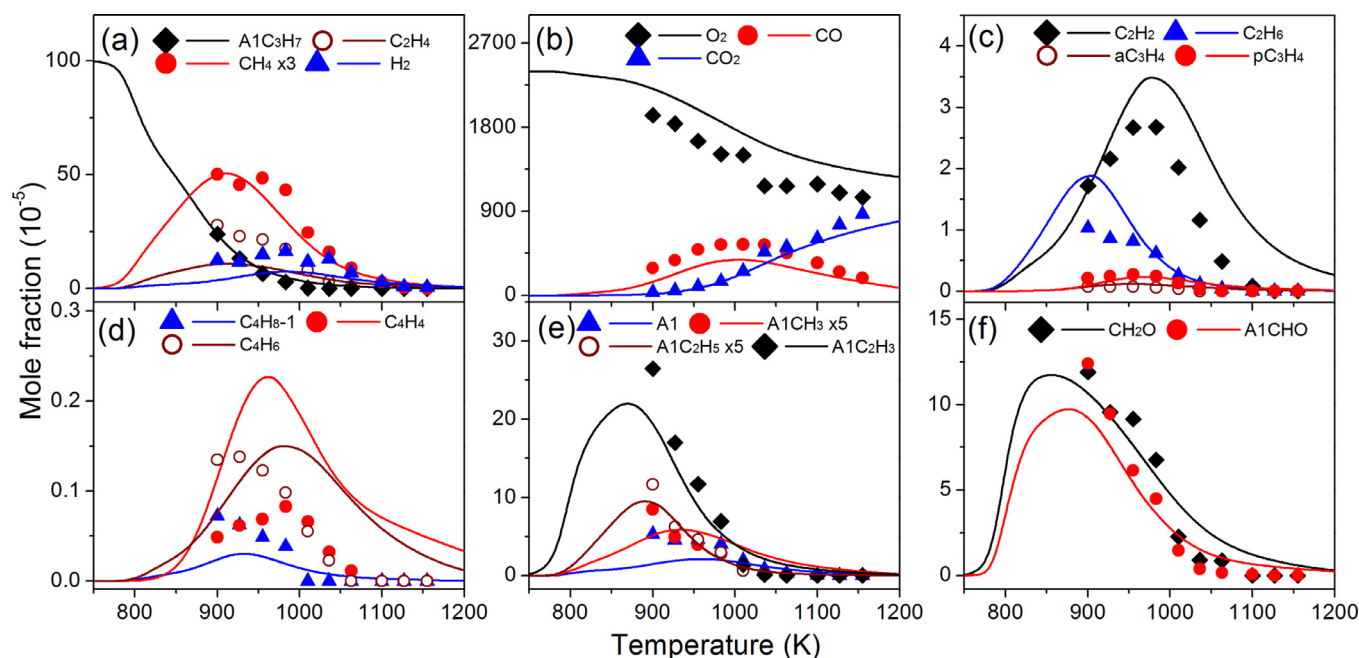


Fig. 5. Experimental (symbols) and simulated (lines) mole fraction profiles of reactants and products during the JSR oxidation of *n*-propylbenzene ($\phi = 0.5$, $P = 10$ atm, $\tau = 0.5$ s).

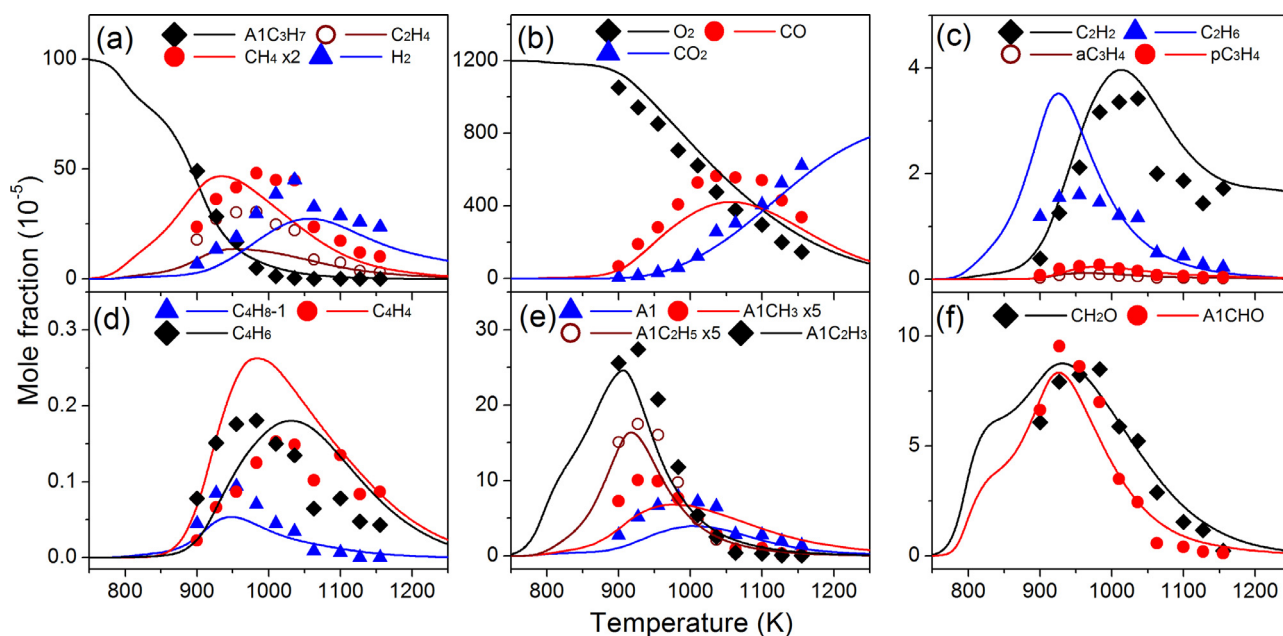


Fig. 6. Experimental (symbols) and simulated (lines) mole fraction profiles of reactants and products during the JSR oxidation of *n*-propylbenzene ($\phi = 1.0$, $P = 10$ atm, $\tau = 0.5$ s).

between the vinylphenyl radical and the methyl radical. Naphthalene can be produced from the benzyl radical via the propargyl addition/cyclization reaction and from styrene via the H-atom abstraction/acetylene addition/cyclization reaction. The former is much more preferred at low pressure while the latter is much more preferred at atmospheric pressure.

4.2. Jet stirred reactor oxidation

Figures 5–8 present the experimental and simulated results in the 10 atm JSR oxidation of *n*-propylbenzene at $\phi = 0.50$, 1.00, 1.50 and 2.00, respectively. The experiments were performed at temperatures over 900 K to investigate the intermediate-to-high

temperature oxidation chemistry of *n*-propylbenzene. The JSR oxidation was simulated using the Perfectly Stirred Reactor module in the Chemkin-Pro software [61]. The ROP analysis was performed at $\phi = 0.50$, 900 K and $\phi = 2.00$, 950 K to provide insights into the oxidation chemistry of *n*-propylbenzene under lean and rich conditions. Figure 9 illustrates the main consumption pathways of *n*-propylbenzene at $\phi = 0.5$ and 2.0. The ROP analysis shows that *n*-propylbenzene is mainly consumed by the H-atom abstraction reactions, while the main H-atom abstraction reactions are different. Under the $\phi = 0.50$ condition, over 70% of *n*-propylbenzene is consumed by the H-atom abstraction reactions by OH, while under the $\phi = 2.0$ condition, the overall contribution of the H-atom abstraction reactions by OH decreases to 40%. Instead, around 30%

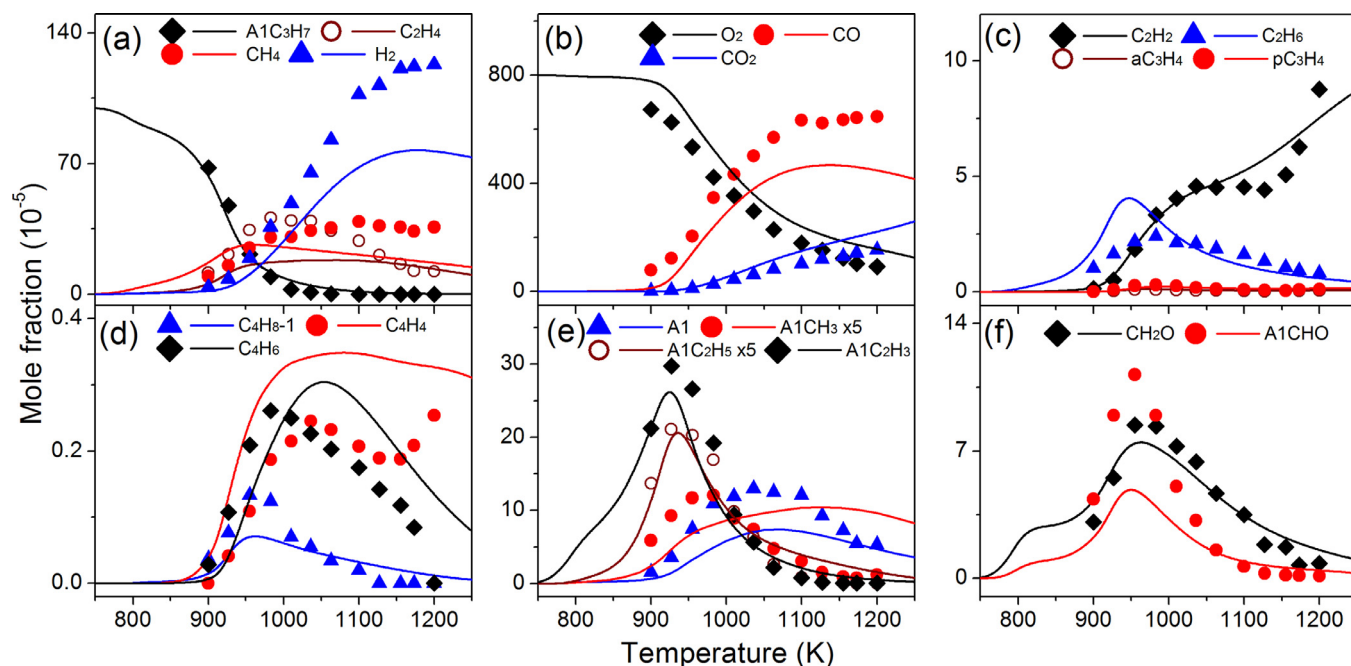


Fig. 7. Experimental (symbols) and simulated (lines) mole fraction profiles of reactants and products in the JSR oxidation of *n*-propylbenzene ($\phi = 1.5$, $P = 10$ atm, $\tau = 0.5$ s).

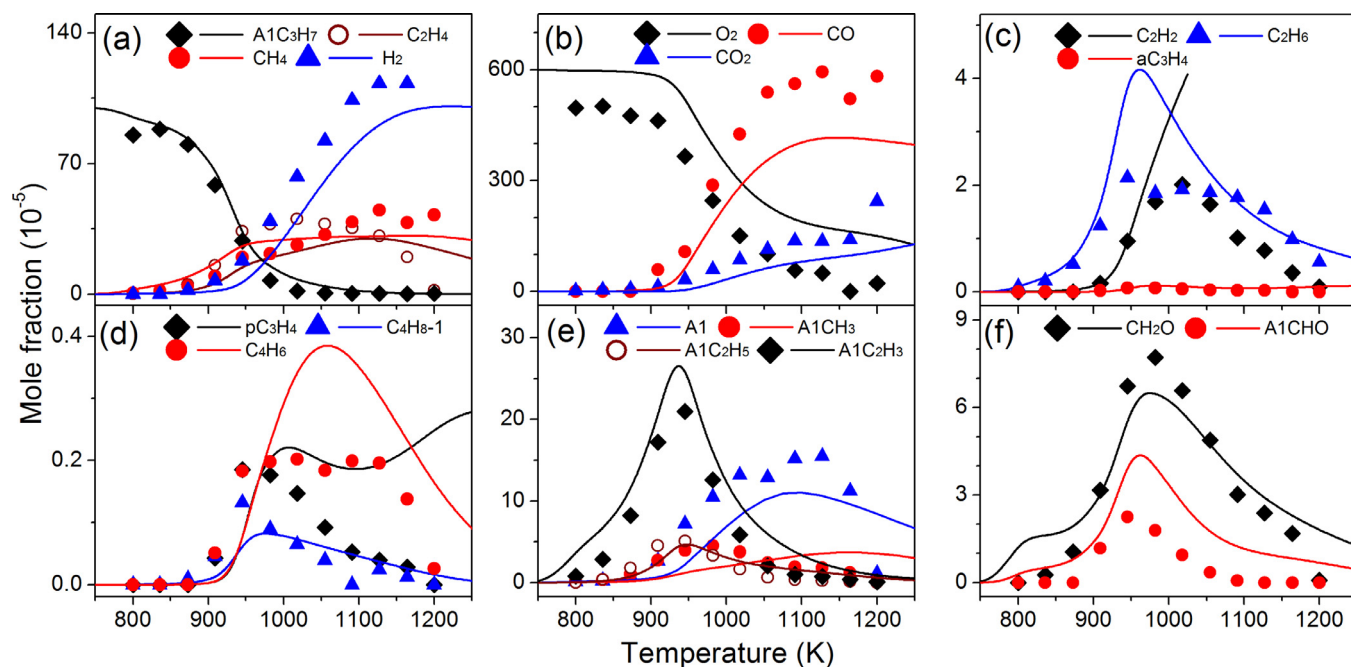


Fig. 8. Experimental (symbols) and simulated (lines) mole fraction profiles of reactants and products in the JSR oxidation of *n*-propylbenzene ($\phi = 2.0$, $P = 10$ atm, $\tau = 0.5$ s).

of *n*-propylbenzene is consumed by the H-atom abstraction reaction by CH_3 under the rich condition. It should be noted that the H-atom abstraction reactions by H atom have only minor contributions to the consumption of *n*-propylbenzene at all equivalence ratios. This is different from toluene and ethylbenzene oxidation under similar conditions [45,46], where H-atom abstraction reactions by H are very important for the consumption of fuel. One possible explanation for this difference can be attributed to the faster formation rate of CH_3 in the oxidation of *n*-propylbenzene since CH_3 can be abundantly produced from the β -C-C scission reaction of phenylpropyl radicals and the oxidation reaction of styrene.

Among the phenylpropyl radicals produced, the 1-phenylpropyl radical is the most favored, due to its resonance stabilized structure as well as the relative weak C-H bond at the benzylic carbon site of *n*-propylbenzene. At all equivalence ratios, the 1-phenylpropyl radical is almost totally decomposed via the β -C-C scission reaction, producing styrene and the methyl radical. This can explain the early and significant production of styrene under the investigated conditions. In contrast, the O_2 -addition reaction has only negligible contributions. This is different from either ethylbenzene oxidation [45] or *n*-butylbenzene oxidation [62] under similar conditions. For example, most of the 1-phenylethyl

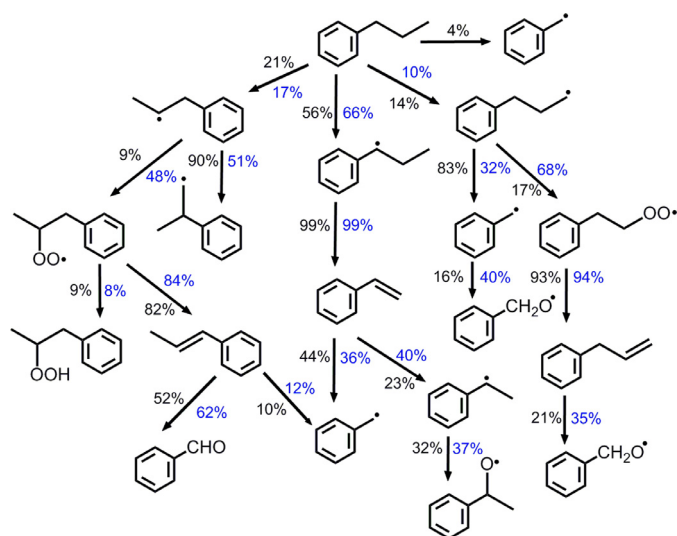


Fig. 9. Main consumption pathways of *n*-propylbenzene in the JSR oxidation of *n*-propylbenzene at 10 atm. The number denotes the contribution of corresponding reaction to the consumption of its major reactant. Blue and black colors represent values at $\phi = 0.5$ and 2.0, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

radical in the oxidation of ethylbenzene at 10 atm is consumed by the O_2 -addition reaction producing the 1-phenylethylperoxyl radical, while in the oxidation of *n*-butylbenzene, around half of the 1-phenylbutyl radical is consumed by the O_2 -addition reaction. The possible reason for this phenomenon is the competition between β -C-C scission reactions and low temperature oxidation reactions. For 1-phenylethyl radical, the β -C-C scission reaction is not available while the β -C-H scission reaction needs a higher barrier, while for 1-phenylbutyl radical, its O_2 -addition reaction product 1-phenylbutylperoxyl radical can readily proceed subsequent low temperature chain branching reactions. In contrast, 1-phenylpropyl radical can easily undergo the β -C-C scission reaction but the low temperature chain branching from its O_2 -addition reaction product 1-phenylpropylperoxyl radical can hardly happen.

Compared with the 1-phenylpropyl radical, the 1-phenylprop-2-yl radical and the 1-phenylprop-3-yl radical are less produced. Besides, their consumption pathways also are dramatically different from those of the 1-phenylpropyl radical. At $\phi = 0.50$, around half of the 1-phenylprop-2-yl radical is consumed by the 1,2-phenyl-migration isomerization reaction producing the 2-phenylpropyl radical, the rest is mainly consumed by O_2 -addition reaction, leading to the production of the 1-phenylprop-2-ylperoxyl radical. As ϕ increases, the contribution of the isomerization reaction increases, making around 90% of 1-phenylprop-2-yl radical consumed by the isomerization reaction at $\phi = 2.0$. 1-Phenylprop-2-ylperoxyl radical mainly converts to 1-phenylpropene via the concerted elimination reaction. For 1-phenylprop-3-yl radical, the O_2 -addition reaction producing the 1-phenylprop-3-ylperoxyl radical and the β -C-C scission reaction are the dominant reactions at $\phi = 0.5$ and 2.0, respectively. Further decomposition of the 1-phenylprop-3-ylperoxyl radical mainly produces 1-phenylprop-2-ene via the concerted elimination reaction.

As seen from Figs. 5–8, ethylbenzene and toluene are abundantly produced in the oxidation of *n*-propylbenzene at all equivalence ratios. This implies that the benzyl radical is one of the most important intermediates in the oxidation of *n*-propylbenzene since both products are mainly formed from the combination reactions of the benzyl radical with alkyl radicals. At $\phi = 0.50$, over 60% of the benzyl radical is formed from the

oxidation reaction of styrene by HO_2 and OH. The rest is mainly produced from the β -C-C scission reaction of the 1-phenyl-3-yl radical and the oxidation reaction of 1-phenylpropene by OH. At $\phi = 2.0$, the main formation pathways of the benzyl radical become pyrolytic reactions, e.g. the β -C-C scission reaction of the 1-phenylprop-3-yl radical and the unimolecular benzylic C-C dissociation reaction of *n*-propylbenzene, which contribute over 55% to the formation of benzyl together. Meanwhile, the oxidation reactions of benzyl and subsequent reactions are the major source of some C_2 – C_6 species, which are similar to those in the oxidation of toluene and ethylbenzene [45,46].

4.3. Laminar premixed flames

In this work, the lean ($\phi = 0.75$) and stoichiometric flames of *n*-propylbenzene were investigated at 0.04 atm using SVUV-PIMS. In addition, Wang et al. [16] investigated the rich laminar premixed flame of *n*-propylbenzene ($\phi = 1.79$) using SVUV-PIMS under the same pressure, argon dilution and cold gas velocity conditions. Therefore, the three laminar premixed flames of *n*-propylbenzene were used to validate the present model. The Premixed Laminar Burner-Stabilized Flame module in the Chemkin-Pro software [61] was used in the simulation. Mixture-averaged transport properties were used and the maximum grids were set as large as possible to ensure an enough refinement. Figure 10 shows the experimental and simulated mole fraction profiles of major flame species including reactants (*n*-propylbenzene and O_2), major products (H_2 , H_2O , CO and CO_2) and Ar. Figures 11 and 12 and S3 and S4 in the Supplementary materials present the experimental and simulated mole fraction profiles of intermediates in the three flames. Based on the ROP analysis, the fuel decomposition and PAH formation processes are discussed below.

The ROP analysis shows that the H-atom abstraction reactions are one of the most important reaction types in the laminar premixed flames of *n*-propylbenzene due to the abundant production of radicals under high temperature oxidation conditions. The H-atom abstraction reactions by OH and H contribute over 90% to the consumption of *n*-propylbenzene in the lean flame. This contribution, however, decreases to around 40% in the rich flame where the unimolecular benzylic C-C dissociation reaction becomes dominant.

Similar to the JSR oxidation, the H-atom abstraction reactions producing resonance stabilized 1-phenylpropyl radical have the largest overall contributions in all three flames, while the 1-phenylpropyl radical mainly converts to styrene and CH_3 via the β -C-C scission reaction. Similarly, the 1-phenylprop-3-yl radical mainly converts to benzyl and ethylene in all flames through the β -C-C scission reactions, respectively. For the 1-phenylprop-2-yl radical, its main consumption pathway is the 1,2-phenyl-migration isomerization reaction producing the 2-phenylpropyl radical. The produced 2-phenylpropyl radical mainly suffers β -C-C scission to form styrene and CH_3 . The further combination reactions of the benzyl radical are the major source of toluene and ethylbenzene. Thus it can be seen from Fig. 11 that benzyl radical, styrene, toluene and ethylbenzene are the major flame intermediates in the laminar premixed flames of *n*-propylbenzene in all flames. The decomposition and oxidation pathways of toluene, ethylbenzene, styrene and benzene are similar to that in the previous studies [45,46,49], which will not be repeated herein.

Phenylpropyl radicals can also undergo O_2 -addition reactions at the low-temperature preheat zone producing the phenylpropylperoxyl radicals, which contribute less than 2% together to the consumption of phenylpropyl radicals. Phenylpropylperoxyl radicals mainly undergo the concerted elimination reactions to produce phenylpropene isomers, which interprets the early formation of phenylpropene isomers in the preheat zone as shown in Fig. 11. In

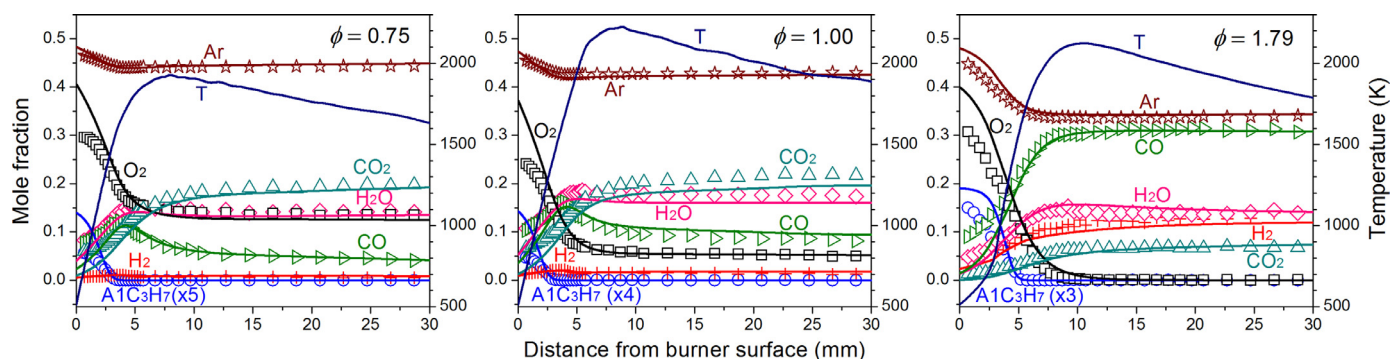


Fig. 10. Experimental (symbols) and simulated (lines) mole fraction profiles of *n*-propylbenzene ($A1C_3H_7$), Ar, O_2 and major products (H_2 , H_2O , CO and CO_2) in the lean, stoichiometric and rich [16] premixed *n*-propylbenzene flames.

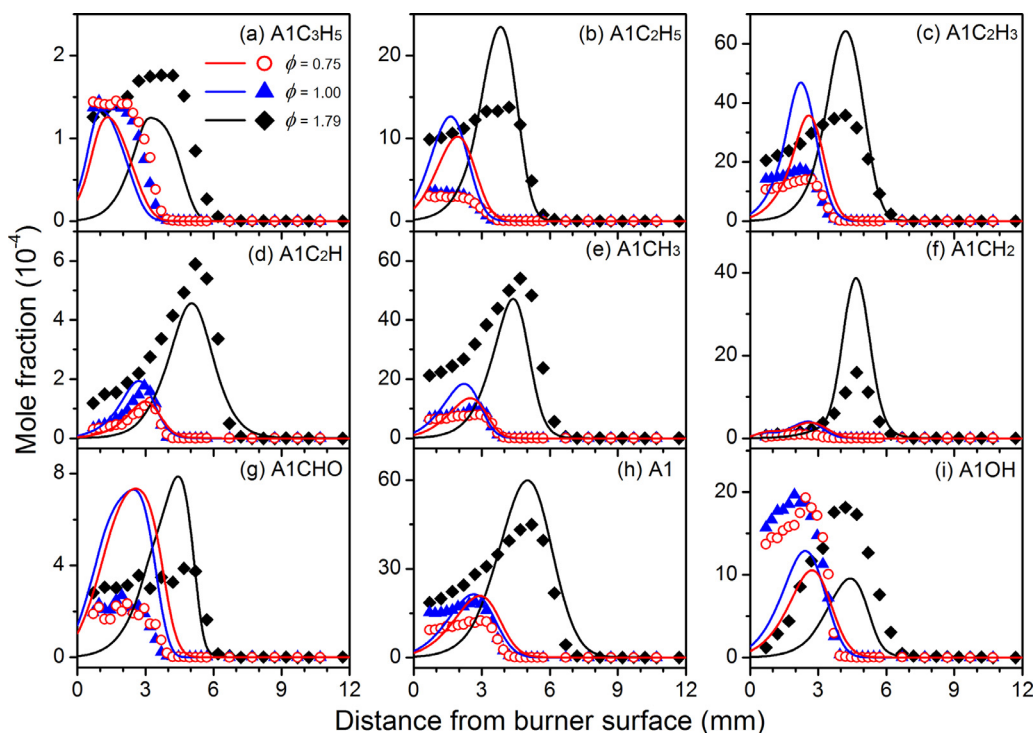


Fig. 11. Experimental (symbols) and simulated (lines) mole fraction profiles of phenylpropene isomers ($A1C_3H_5$), ethylbenzene ($A1C_2H_5$), styrene ($A1C_2H_3$), phenylacetylene ($A1C_2H$), toluene ($A1CH_3$), benzyl radical ($A1CH_2$), benzaldehyde ($A1CHO$), benzene ($A1$) and phenol ($A1OH$) in the lean, stoichiometric and rich [16] premixed *n*-propylbenzene flames.

the flame zone, phenylpropene isomers can also be produced from the β -C–H scission reactions of phenylpropyl radicals. The consumption of phenylpropene isomers can produce indene, benzaldehyde ($A1CHO$), benzoyl radical ($A1CO$), benzene and phenyl radical.

Figure 12 shows the experimental and simulated mole fraction profiles of major PAH species. Indene, naphthalene, acenaphthylene ($A2R5$) and diphenylmethane are the abundantly produced PAHs under the flame conditions. The maximum mole fractions of non-fused PAHs such as biphenyl and diphenylmethane are smaller than those of indene and naphthalene, which is different from the pyrolysis. In particular, indene is the most abundantly produced PAH under the flame conditions, even in the lean and stoichiometric flames, while its concentration levels in the pyrolysis is several times lower than biphenyl and diphenylmethane. In all flames, indene is mainly formed from the reaction between the vinylphenyl radical and the methyl radical. Other important formation pathways of indene include the cyclization of the phenylpropenyl radical, the H-addition reaction of the indenyl radical (C_9H_7) and the decomposition of naphthalenols. Among them, the cyclization

reaction of the phenylpropenyl radical is the most significant one, especially in the preheat zone of the flame. Naphthalene is also an important PAH in the flames of *n*-propylbenzene. Due to the abundant production of styrene in the flames, the acetylene addition reaction of the phenylvinyl radical is the major formation pathway of naphthalene. Other important formation pathways include the self-combination reaction of the cyclopentadienyl radical, the reaction between the phenyl radical and vinylacetylene, and the isomerization reaction of methyleneindene. For phenanthrene, its formation pathways are very similar to those in the flames of toluene [46], e.g. self-combination reaction of the fulvenallenyl radical, cyclization reaction of the stilbenyl radical, cyclization reaction of the hydro-methylenefluorene radical, phenyl addition/cyclization reaction of the acetylphenyl radical and so on.

4.4. Model validation against literature data

Besides the pyrolysis, oxidation and flame data reported in this work, a wide range of literature data were also used to validate the

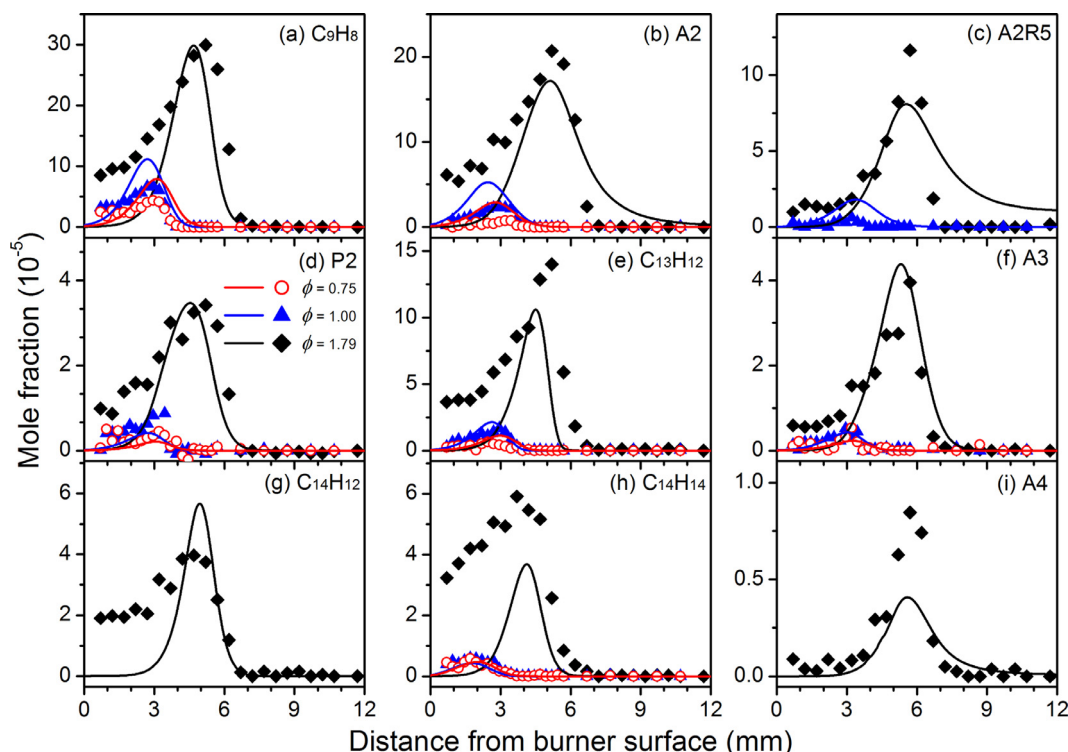


Fig. 12. Experimental (symbols) and simulated (lines) mole fraction profiles of indene (C_9H_8), naphthalene (A2), acenaphthylene (A2R5), biphenyl (P2), diphenylmethane ($C_{13}H_{12}$), phenanthrene (A3), stilbene ($C_{14}H_{12}$), bibenzyl ($C_{14}H_{14}$) and pyrene (A4) in the lean, stoichiometric and rich [16] premixed *n*-propylbenzene flames.

Table 3

Summary of the *n*-propylbenzene combustion data used to validate the present model.

Experimental type	Temperature (K)	Pressure (atm)	Equivalence ratio	Reference
Flow reactor pyrolysis	750–1400	0.04, 0.2, 1	∞	This work
Shock tube pyrolysis	1000–1700	50	∞	[15]
Jet-stirred reactor oxidation	750–1200	10	0.5, 1.0, 1.5, 2.0	This work
	850–1250	1	0.5, 1.0, 1.5	[13]
Shock tube oxidation	900–1550	28, 24	0.54, 1.90	[14]
	850–1650	51, 49, 52	0.55, 1.00, 1.90	[14]
Flow reactor oxidation	1060	1	0.65, 1.00, 1.50	[11]
Laminar premixed flame	$T_{max} = 2200$	0.04	0.75, 1.00	This work
		0.04	1.79	[16]
Ignition delay times	800–1450	1, 10, 30, 5	0.29, 0.48, 0.96, 1.92	[18,19]
	950–1330	11, 21	0.5, 1.0, 1.5	[63]
Laminar flame speeds	$T_u = 358, 398$	1	0.55–1.60	[23]
	$T_u = 400$	1	0.7–1.4	[20, 21, 64]
	$T_u = 400$	1–3	0.7–1.4	[21]
	$T_u = 470$	1	0.7–1.4	[21]
	$T_u = 353$	1	0.7–1.5	[22,64]
Extinction strain rates	$T_{fuel} \text{ (inlet)} = 400$	1	$X_{fuel} = 0.09\text{--}0.20$	[24]
	$T_u = 353$	1	Fuel/ N_2 mass ratio = 0.08–0.10	[22]
	$T_u = 400$	1	0.8–1.6	[20]

present model, including the atmospheric pressure JSR oxidation, atmospheric pressure flow reactor oxidation, high pressure shock tube pyrolysis and oxidation, ignition delay times, laminar flame speeds, and extinction strain rates. Table 3 summarizes all the experimental data used for the validation of the present model.

4.4.1. Atmospheric pressure JSR and flow reactor oxidation

Dagaut et al. [13] investigated the oxidation of *n*-propylbenzene in a jet stirred reactor at 1 atm and equivalence ratios of 0.5, 1.0 and 1.5. These data were used to validate the present model. Figure 13 illustrates experimental and simulated mole fraction profiles of *n*-propylbenzene and oxidation products at $\phi = 1.0$. The comparisons of experimental and simulated results at $\phi = 0.5$ and 1.5 are presented in Figs. S5 and S6 in the Supplementary materials, respectively.

The main consumption pathways of *n*-propylbenzene at 1 atm are slightly different from those at 10 atm. Over 35% of *n*-propylbenzene is consumed by benzylic C–C bond dissociation reaction, producing the benzyl radical and the ethyl radical. The rest is consumed via the H-atom abstraction reactions by OH, H, O and CH_3 , forming phenylpropyl radicals. The consumption of 1-phenylpropyl and 1-phenyl-prop-3-yl radicals mainly proceeds via the β -C–C scission reactions, while the consumption of the 1-phenyl-prop-2-yl radical is mainly controlled by the phenyl-migration isomerization reaction.

The present model was also validated against the atmospheric pressure flow reactor oxidation data reported by Litzinger et al. [11]. The experiments were performed at a temperature near 1060 K and equivalence ratios of 0.65, 1.0, and 1.5. The comparisons of experimental and simulated mole fraction profiles of

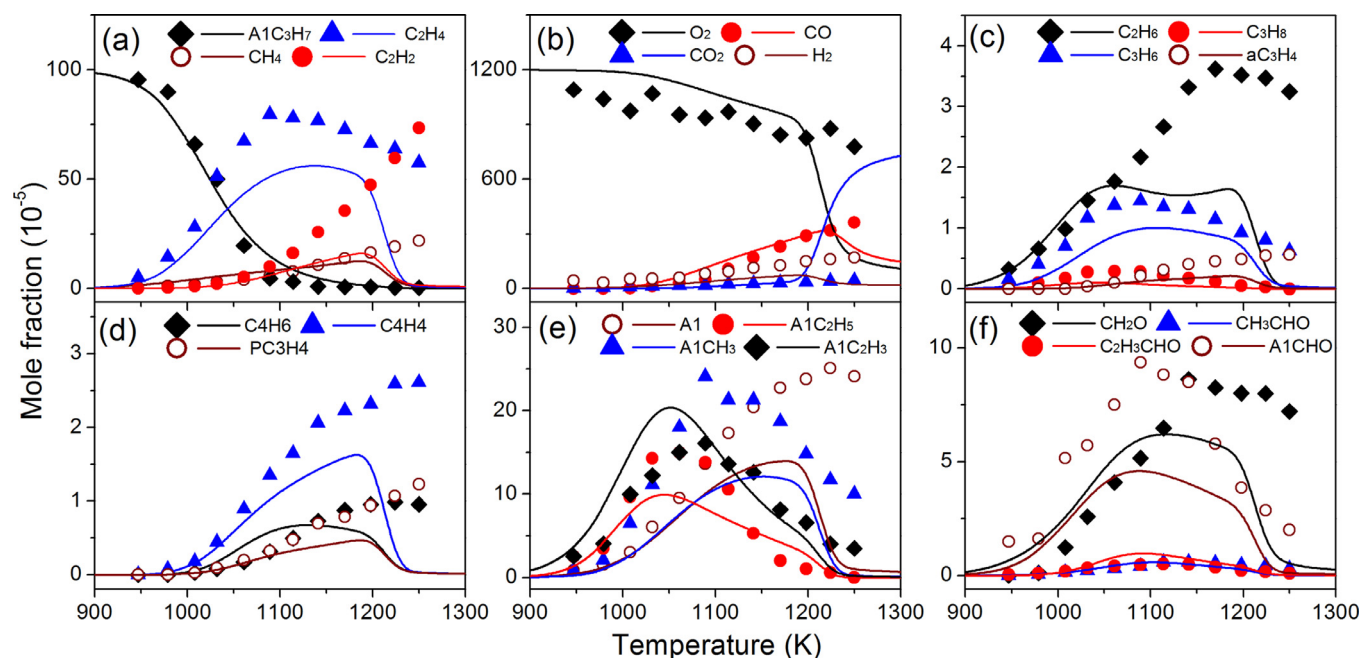


Fig. 13. Validation against the mole fraction profiles of reactants and products in the JSR oxidation of *n*-propylbenzene ($\phi = 1.0$, $P = 1$ atm, $\tau = 0.07$ s) [13]. Symbols are the literature data and lines are the simulated results of the present model.

oxidation species are shown in Figs. S7–S9 in the *Supplementary materials*. In general, the present model can reasonably reproduce the decomposition of fuel and the formation of oxidation species.

4.4.2. High pressure shock tube pyrolysis and oxidation

Gudiyella and Brezinsky investigated the high pressure pyrolysis [15] and oxidation [14] of *n*-propylbenzene in the shock tube. The pyrolysis experiments were performed at 50 atm and 1027–1678 K, the oxidation experiments were performed at pressures of 25 and 50 atm, at temperatures from 838 to 1669 K and at equivalence ratios of 0.5–1.9. These high pressure pyrolysis and oxidation data were also used to validate the present model. Figure 14 illustrates the experimental and simulated mole fraction profiles of *n*-propylbenzene and products in the shock tube pyrolysis of *n*-propylbenzene at 50 atm. The present model can well reproduce the decomposition of *n*-propylbenzene and formation of major products below ~ 1450 K, while at higher temperatures, the model under-predicts the consumption of toluene, benzene, ethylene and methane. Figure 15 presents the experimental and simulated mole fraction profiles in the shock tube oxidation of *n*-propylbenzene at 51 atm and $\phi = 0.55$. The comparisons under other oxidation conditions are presented in Figs. S10–S13 in the *Supplementary materials*.

Under the high pressure pyrolysis conditions, the decomposition of *n*-propylbenzene is mainly controlled by the benzylic C–C bond dissociation reaction (54%) and H-atom abstraction reactions by H (32%), which is very similar to that in the flow reactor pyrolysis of *n*-propylbenzene at 1 atm. Under the high pressure oxidation conditions, the benzylic C–C bond dissociation reaction is still the major decomposition reaction of *n*-propylbenzene and contributes over 35% to the consumption of *n*-propylbenzene. The rest of *n*-propylbenzene is consumed via the H-atom abstraction reaction by OH, H, O and CH₃, forming phenylpropyl radicals. The consumption of the benzyl radical in the high pressure shock tube oxidation is very similar to that in the high pressure JSR oxidation, thus will not be repeated herein. The main decomposition pathways of phenylpropyl radicals are similar to those under the pyrolysis condition, i.e. the β -C–C scission reactions and

phenyl-migration isomerization reaction, while the O₂-addition reaction has almost no contribution due to the short residence time in the shock tube oxidation.

4.4.3. Ignition delay times

Darcy et al. [18] measured the ignition delay time of *n*-propylbenzene in a shock tube at equivalence ratios of 0.29, 0.48, 0.96 and 1.92, at pressures of 1, 10 and 30 atm and at temperatures from 1000–1600 K. Then they [19] extended the shock tube measurements to the pressure of 50 atm and covered the temperature range of 650–1000 K by using a rapid compression machine. Wang et al. [63] measured the ignition delay time of *n*-propylbenzene in a shock tube at equivalence ratios of 0.5, 1.0 and 1.5, at pressures of about 11 and 21 atm, and at temperatures from 809 to 1364 K. Figure 16 compares the simulated results by the present model with the experimental results reported by Darcy et al. [18,19] and Wang et al. [63]. The simulation was performed with the Closed Homogeneous Batch Reactor module in the Chemkin-Pro software [61]. The shock tube was treated as a zero-dimensional adiabatic reactor with constant volume. The simulated ignition delay time was defined as the time at which the OH concentration reached half of its maximum, and is also very close to the time of maximum temperature rising. The simulation of rapid compression machine used volume profiles generated from non-reactive pressure traces, which were provided with the experimental data [19].

ROP analysis and sensitivity analysis were performed at $\phi = 0.96$, $P = 50$ atm and $T = 800$ K to determine the most significant reaction for the low temperature ignition process of *n*-propylbenzene. The results of sensitivity analysis are presented in Fig. 17, while the results of ROP analysis are illustrated in Fig. S14 in the *Supplementary materials*. The sensitivity analysis at 800 K indicates that the H-atom abstraction reaction of *n*-propylbenzene by HO₂ producing 1-phenylpropyl radical and H₂O₂ has the largest sensitivity coefficient to reduce the ignition delay time of *n*-propylbenzene. Other reactions that will promote the oxidation reactivity of *n*-propylbenzene include the decomposition reaction of H₂O₂ to two OH, H-atom abstraction reaction of *n*-propylbenzene by methylperoxy (CH₃OO) radical and the oxidation reactions of styrene by HO₂. Some key reactions in the low temperature

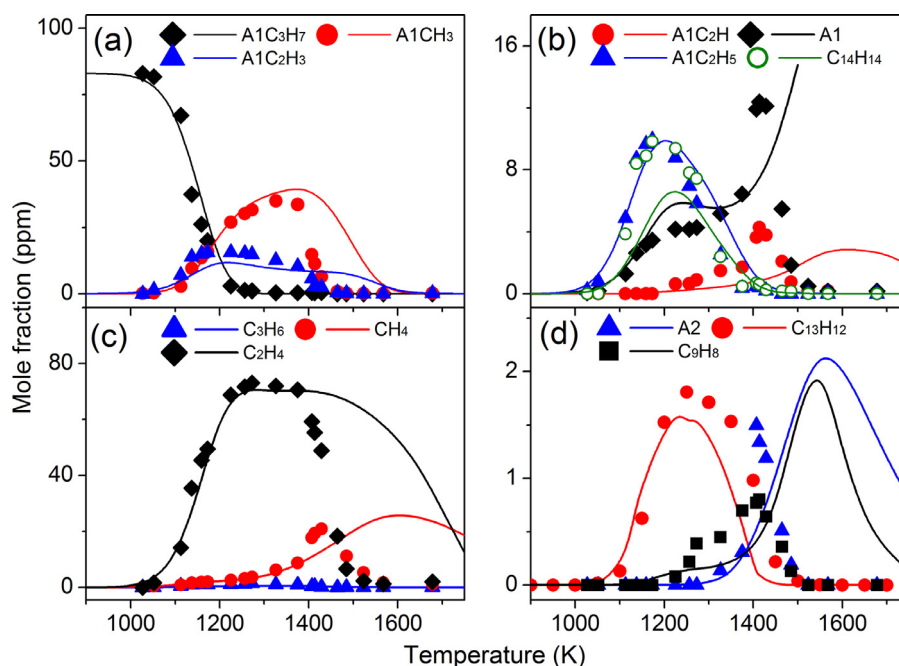


Fig. 14. Validation against the mole fraction profiles of reactant and pyrolysis products in the shock tube pyrolysis of *n*-propylbenzene ($P = 50$ atm, $\tau = 1.6$ – 2.0 ms) [15]. Symbols are the literature data and lines are the simulated results of the present model.

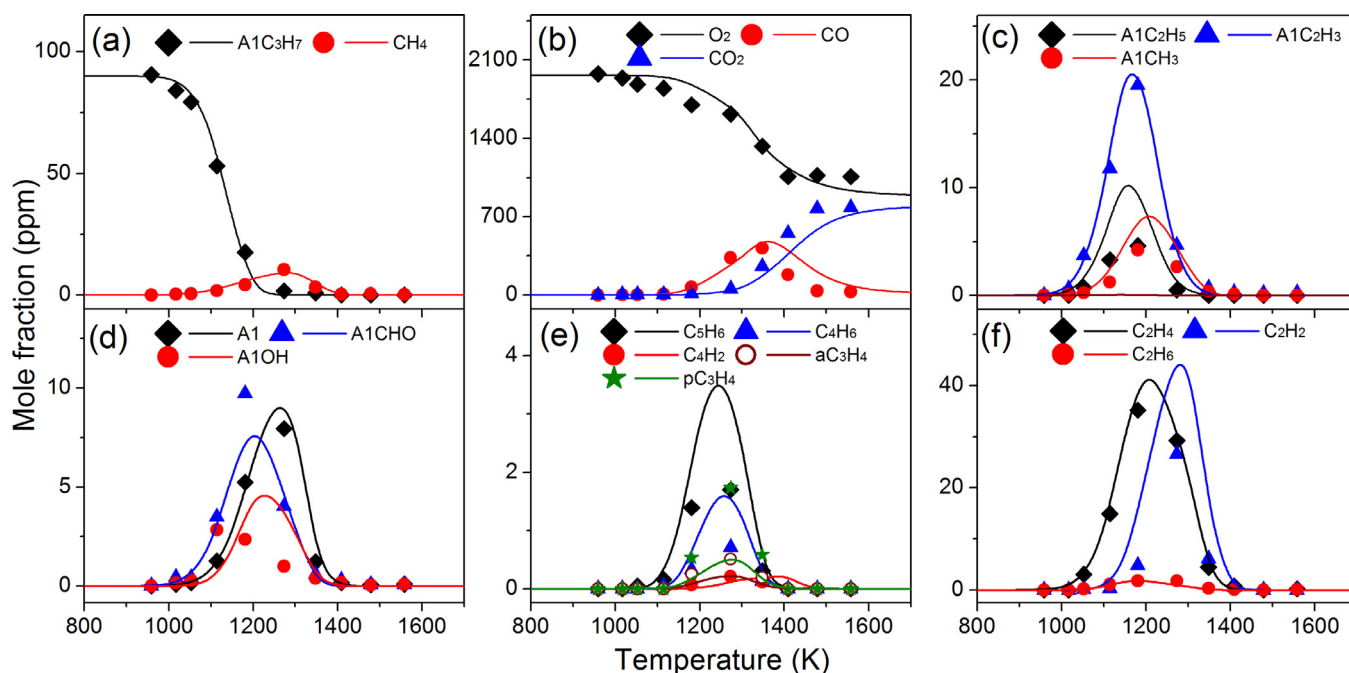


Fig. 15. Validation against the mole fraction profiles of reactant and oxidation products in the shock tube oxidation of *n*-propylbenzene ($P = 51$ atm, $\phi = 0.55$, $\tau = 1.27$ – 1.78 ms) [14]. Symbols are the literature data and lines are the simulated results of the present model.

chain branching process also have large sensitivity coefficients to promote the oxidation reactivity of *n*-propylbenzene, e.g. the decomposition reaction of OOQOOH to ketohydroperoxide and O_2 -addition reaction of QOOH . Important reactions that decrease the oxidation reactivity of *n*-propylbenzene include the chain termination reaction $\text{HO}_2 + \text{HO}_2 = \text{H}_2\text{O}_2 + \text{O}_2$, $\text{CH}_3\text{OO} + \text{HO}_2 = \text{CH}_3\text{OOH} + \text{O}_2$ and concerted elimination reactions of phenylpropylperoxides.

4.4.4. Laminar flame speeds

Kim et al. [64] and Hui et al. [20] both measured the laminar flame speeds of *n*-propylbenzene at atmospheric pressure and the

initial temperature of 400 K. Hui and Sung [21] measured the laminar flame speeds of *n*-propylbenzene at 1, 2 and 3 atm and the initial temperature of 400 K, as well as at atmospheric pressure and the initial temperature of 470 K. Ji et al. [22] measured the laminar flame speeds of *n*-propylbenzene at atmospheric pressure and the initial temperature of 353 K. Mehl et al. [23] measured the laminar flame speeds of *n*-propylbenzene at atmospheric pressure and the initial temperatures of 358 K and 398 K. The present model was validated against these laminar flame speed data, as shown in Fig. 18. The simulation was performed using the Premixed Laminar Flame-speed Calculation module in the Chemkin-Pro software [61]. The thermal diffusion effects were

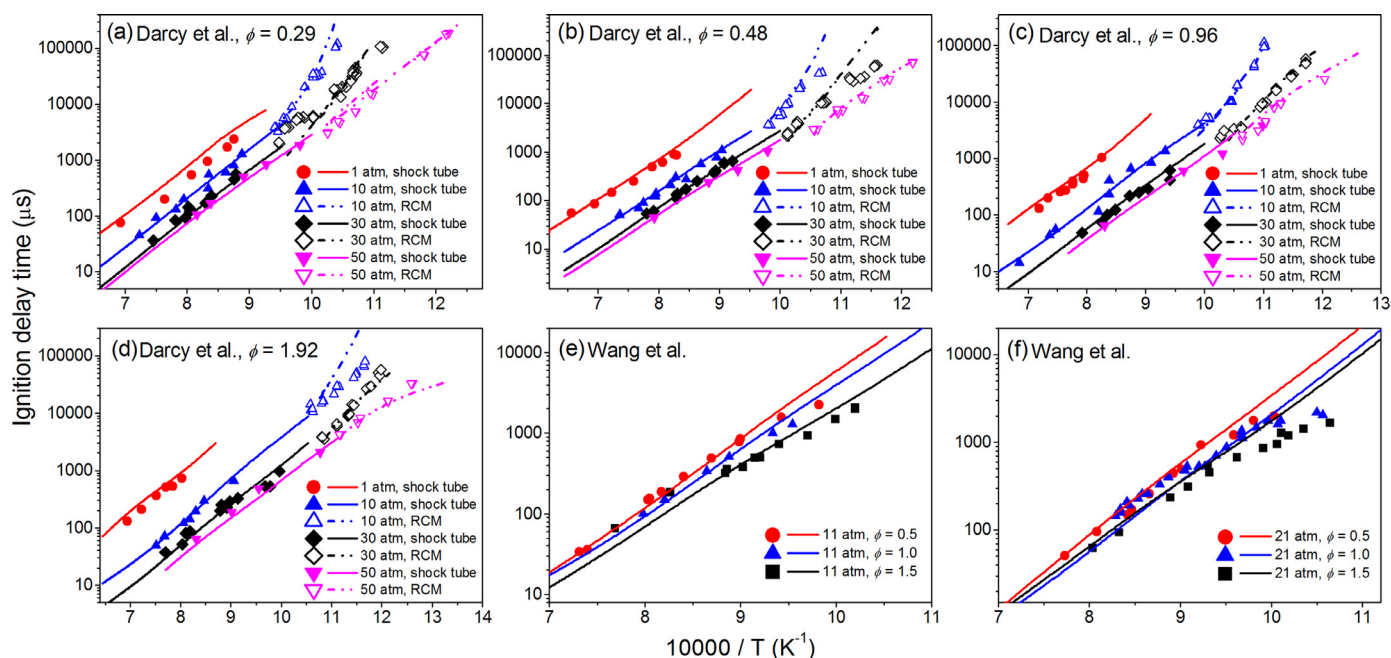


Fig. 16. Validation against the ignition delay times of *n*-propylbenzene at a variety of pressures, temperatures and equivalence ratios reported by (a)–(d) Darcy et al. [18,19] and (e,f) Wang et al. [63]. Symbols are the experimental results and lines are the simulated results of the present model.

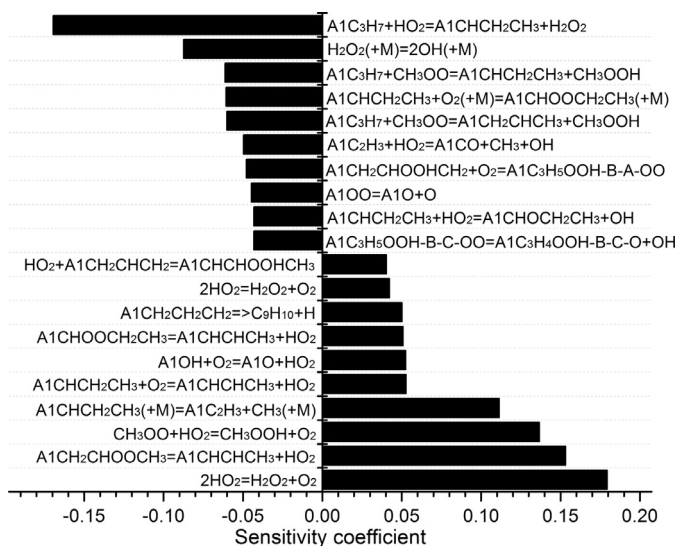


Fig. 17. Sensitivity analysis of the ignition delay time of *n*-propylbenzene at $\phi = 0.96$, $P = 50$ atm and $T = 800$ K.

considered and the mixture-averaged transport properties were used.

4.4.5. Extinction strain rates

Extinction strain rate data taken from the literature were also used to validate the present model. Won et al. [24] measured the extinction strain rates of *n*-propylbenzene in counterflow diffusion flames with different fuel mole fraction. Ji et al. [22] investigated the extinction strain rates of *n*-propylbenzene in counterflow diffusion flames with different fuel/N₂ mass ratio. Hui et al. [20] measured the extinction strain rates of *n*-propylbenzene in the counterflow premixed flames with different equivalence ratio. Figure 19 shows the comparison of experimental and simulated results of the extinction strain rate data. The simulation was performed using the OPPDIF code of the Chemkin II program [65] with the arch-length continuation method.

5. Conclusions

In this work, flow reactor pyrolysis, jet stirred reactor oxidation, and laminar premixed flames of *n*-propylbenzene were

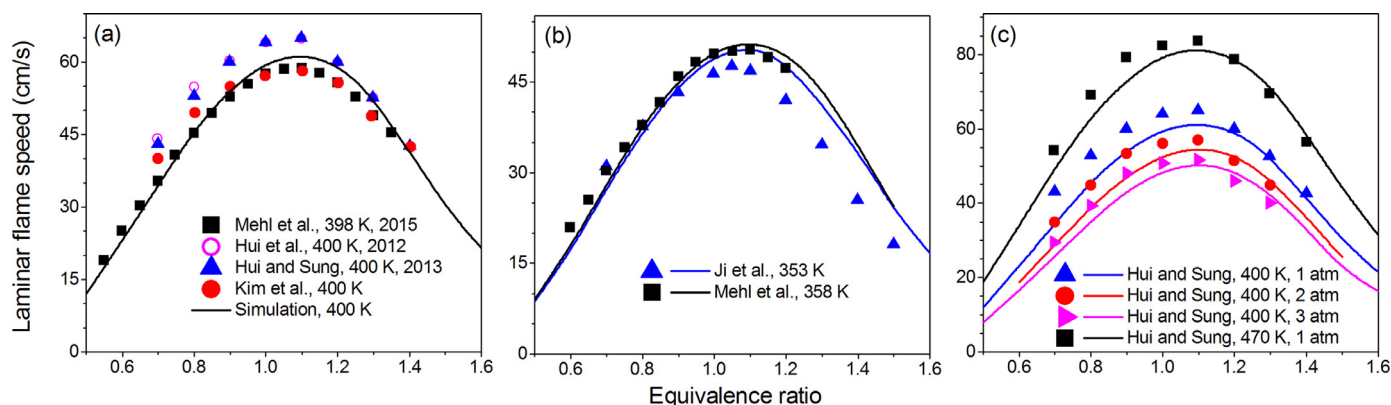


Fig. 18. Validation against the laminar flame speeds of *n*-propylbenzene at different pressures and initial temperatures [20–23, 64]. Symbols are the experimental results and lines are the simulated results using the present model.

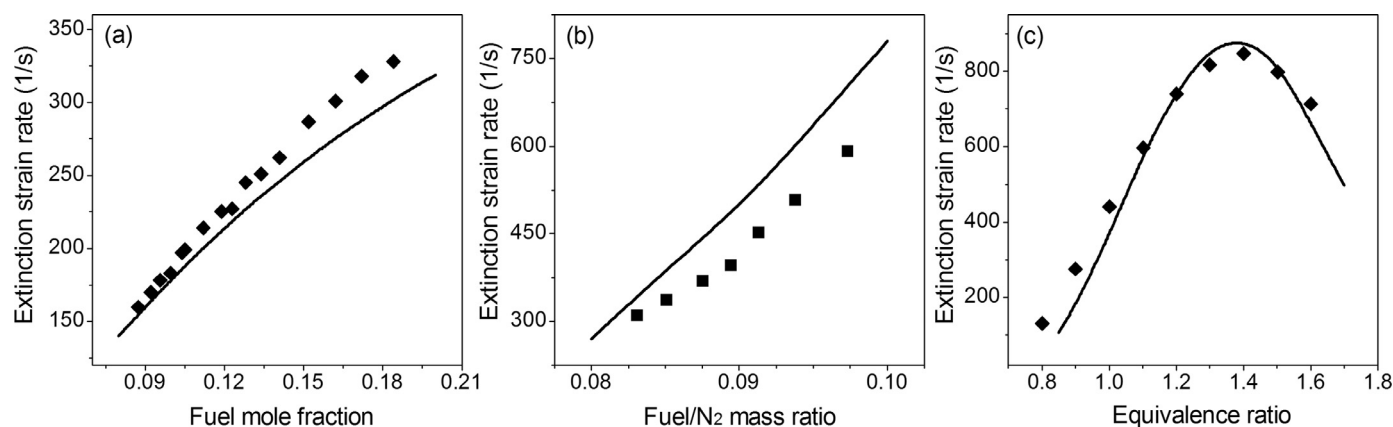


Fig. 19. Validation against the extinction strain rates of *n*-propylbenzene reported by (a) Won et al. [24] and (b) Ji et al. [22] in counterflow diffusion flames and (c) Hui et al. [20] in counterflow premixed flames.

investigated. A detailed kinetic model for *n*-propylbenzene combustion was developed based on our previous kinetic model for ethylbenzene. In the flow reactor pyrolysis of *n*-propylbenzene, the benzylic C–C bond dissociation reaction is the dominant reaction for *n*-propylbenzene decomposition at all pressures, while the contribution of H-atom abstraction reaction increases with increasing pressure. Compared with ethylbenzene, *n*-propylbenzene has higher thermal decomposition rate due to its weaker benzylic C–C bond and the additional alkyl C–C bond. In the jet stirred reactor oxidation at 10 atm, *n*-propylbenzene is mainly consumed via the H-atom abstraction reactions by radical attack. Low temperature oxidation reactions also play important roles under the investigated conditions. The H-atom abstraction by radical attack is the most important reaction type in the low-pressure laminar premixed flames of *n*-propylbenzene, especially in the lean flame. As the equivalence ratio increases, the role of benzylic C–C bond dissociation reaction is enhanced. Besides the present data, the kinetic model was also validated against the combustion data of *n*-propylbenzene in literature. Good agreements can be observed between the experimental and simulated results, demonstrating the ability of the present model to predict *n*-propylbenzene combustion over a wide range of pressure, temperature and equivalence ratio conditions.

Acknowledgments

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.combustflame.2017.08.010](https://doi.org/10.1016/j.combustflame.2017.08.010).

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